

```
student_name = "Gordon" # fill your name
student_id = "s223532749" # fill your student ID
print("Student name: " + student_name)
print("Student ID: " + student_id)
```

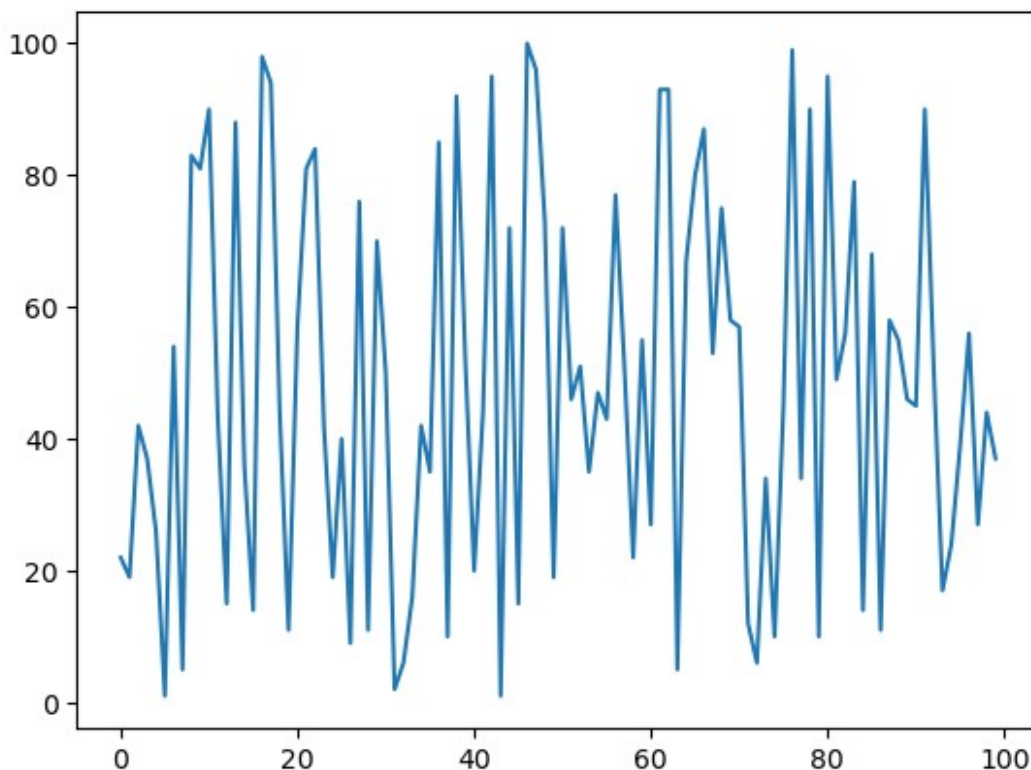
Student name: Gordon
Student ID: s223532749

```
import random
import matplotlib.pyplot as plt
```

```
n_values = 100
y_values = []
```

```
# Create data (y_values) randomly between 1 and 100.
for i in range(n_values):
    y_values.append(random.randint(1, 100))
```

```
x_values = range(n_values) # X is sequence of values 0-99
plt.plot(x_values, y_values)
plt.show()
```



```
# Plot 2 variables
#
```

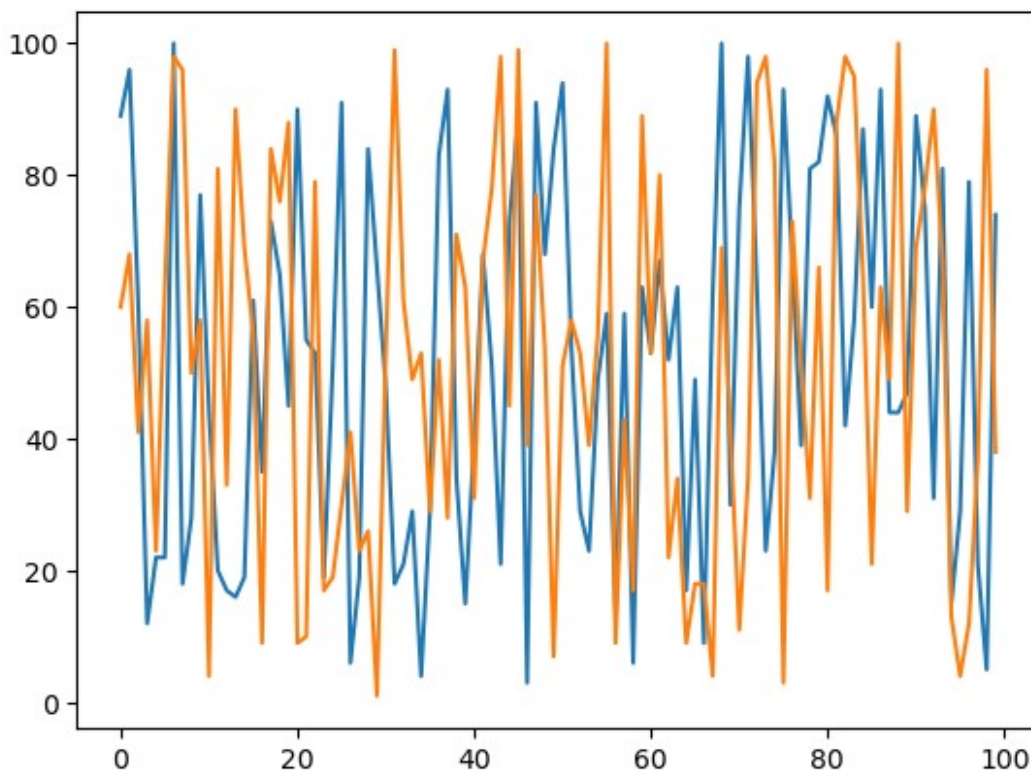
```

n_values = 100
y_values_1 = []
y_values_2 = []

# Create data (y_values) randomly between 1 and 100.
for i in range(n_values):
    y_values_1.append(random.randint(1, 100))
    y_values_2.append(random.randint(1, 100))

x_values = range(n_values) # X is sequence of values 0-99
plt.plot(x_values, y_values_1)
plt.plot(x_values, y_values_2) # call plot again draws in the same
graph.
plt.show()

```

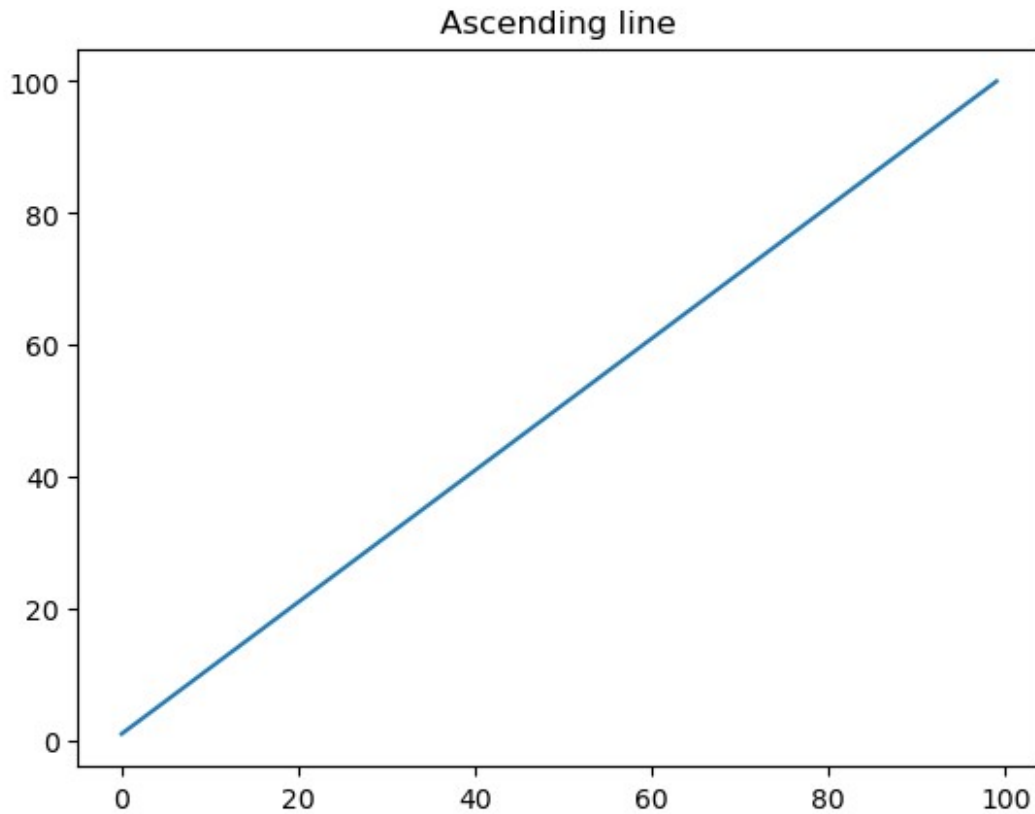


```

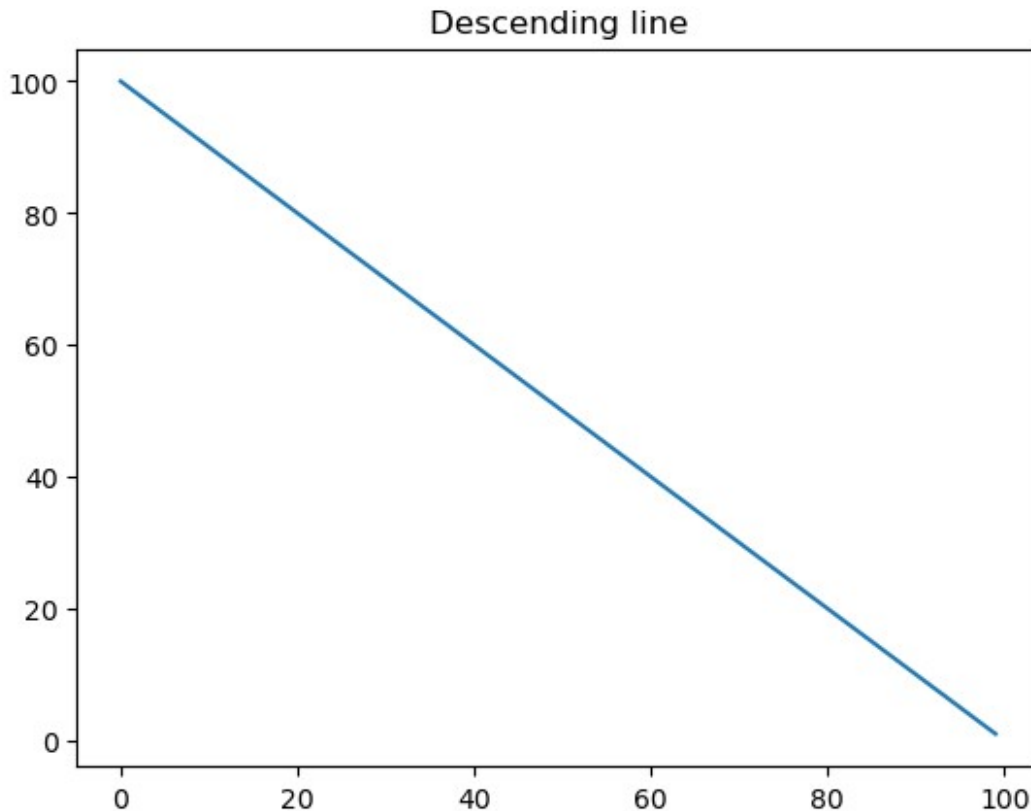
#
# Activity 1: Create data so that the plot draws an
# ascending line (y_values increase at any rate).
#

y_values_asc = list(range(1, n_values + 1))
plt.plot(x_values, y_values_asc)
plt.title("Ascending line")
plt.show()

```



```
#  
# Activity 2: Create data so that the plot draws a  
# descending line (y_values decrease at any rate).  
#  
y_values_desc = list(range(n_values, 0, -1))  
plt.plot(x_values, y_values_desc)  
plt.title("Descending line")  
plt.show()
```



```
#  
# Activity 3: Create data so that the plot draws a  
# wave. You can consider using Python's math library, which has  
# a sin function (detail  
https://www.w3schools.com/python/ref\_math\_sin.asp).  
#  
import math  
  
amplitude = 50  
cycles = 3 # number of full sine cycles across the x range  
y_wave = [amplitude * math.sin(2 * math.pi * cycles * (i / n_values))  
for i in x_values]  
  
plt.plot(x_values, y_wave)  
plt.title("Sine wave")  
plt.show()
```

Sine wave

